

Ubuntu Dual-Boot Install

Do this over the weekend, before Week 2. Bring problems to class or office hours — don't struggle alone.

Step 1: Download the ISO

<https://mirrors.sustech.edu.cn/ubuntu-releases/22.04/ubuntu-22.04.5-desktop-amd64.iso> (4.4GB)

Step 2: Burn a USB drive

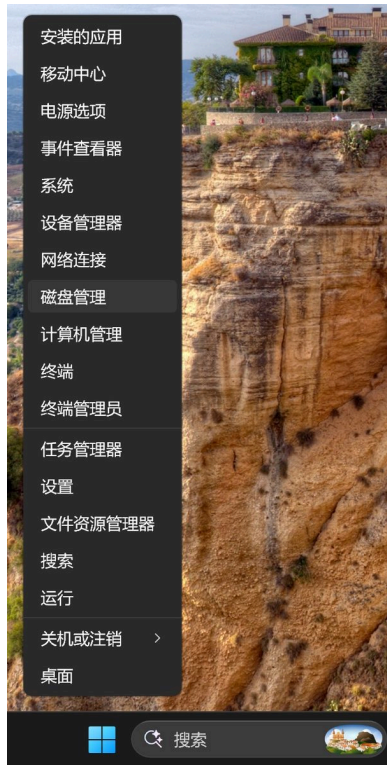
- 1 Install [Rufus](#)
- 2 Select the ISO you downloaded + your USB drive
- 3 Click burn, wait (10-20 min — start this ahead of time, don't wait around on it)

Single-disk laptop: you need a USB drive. Two-disk laptop (two physical drives): put the ISO on the other drive instead, no USB needed.

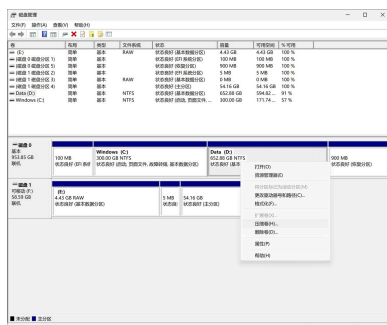
Step 3: Shrink a 256GB partition

Minimum 64GB. 256GB is safer (fits ROS2 + WeChat + VSCode + NoMachine + conda comfortably).

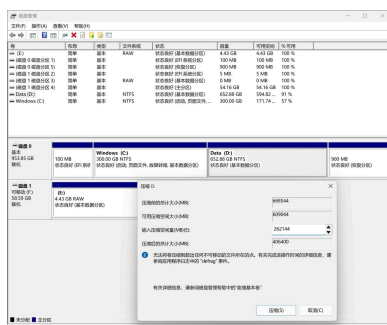
- 1 Right-click the Windows icon (bottom-left) → Disk Management



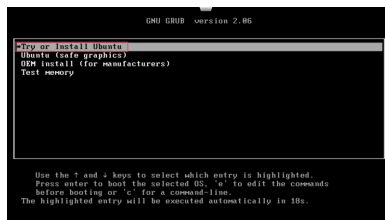
2 Right-click any partition (doesn't have to be C:, same physical disk is fine) → Shrink Volume



3 Enter 262144 (256GB, 1GB=1024MB) → Shrink



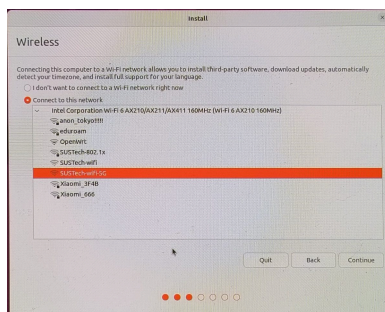
4 Leave the new “Unallocated” space alone



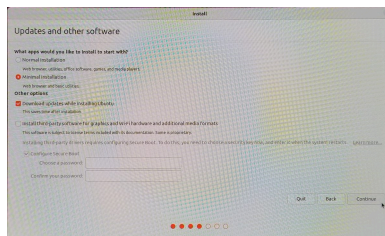
(Source: dev.to)

Step 5: Partition (choose “Something else” in the installer)

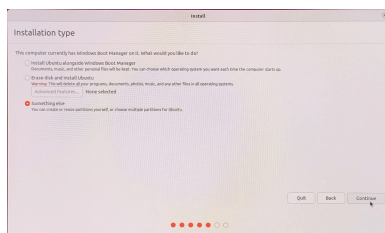
- 1 Wi-Fi: connect to **SUSTech-wifi-5G**, no password needed. Skip it if it fails — you can connect after install



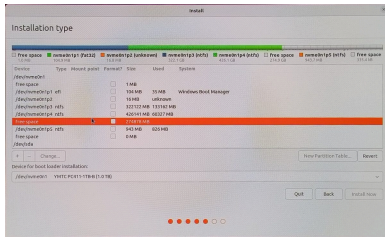
- 2 Updates and other software → Minimal installation



- 3 Installation type → “Something else”



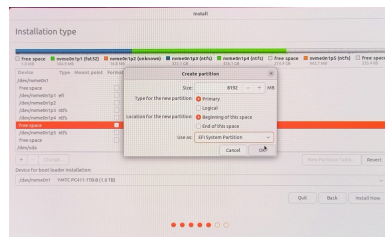
- 4 Find the “free space” block you just freed up in Windows (units are MB here — the number won’t match the 256GB from Windows exactly, that’s normal, just 1024-based rounding)



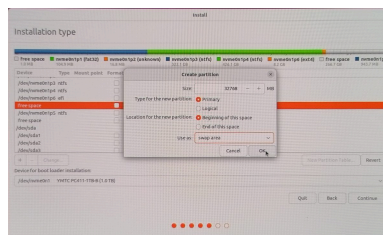
- 5 Select it, click “+” bottom-left, create three partitions, all “Primary” (skip a separate `/home`, not needed):

Partition	Size	Mount point
EFI	100MB	<code>/boot/efi</code>
swap	2GB	swap
root	rest of the space	<code>/</code>

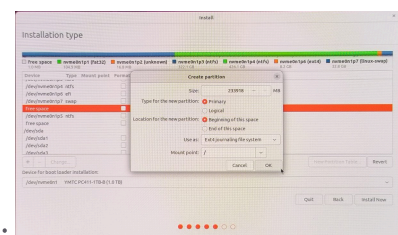
Screenshots below are from a different machine — numbers won’t match the table, just follow the steps:



Use as “EFI System Partition” :

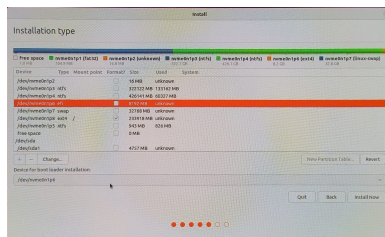


Use as “swap area” :

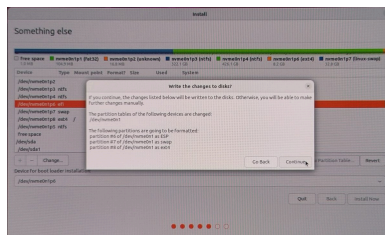


Use as “Ext4 journaling file system” , Mount point `/`:

- 6 “Device for boot loader installation” at the bottom → pick the EFI partition you just created (screenshot below picked `/dev/nvme0n1p6` — yours will differ, just don’t pick the whole disk)

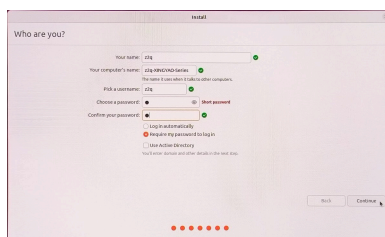


- 7 Install Now → check the partitions in the confirmation dialog → Continue

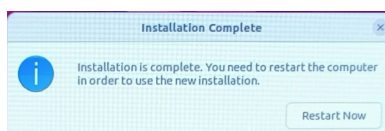


Step 6: Create your account, reboot

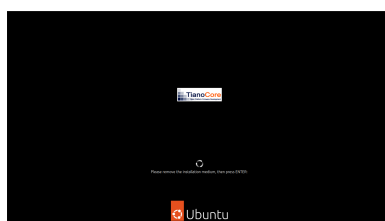
- 1 Fill in name / computer name / username / password — anything works, just remember the password. Don't overthink it — you'll type it constantly (`sudo` needs it every time). Our lab robots' password is a single space.



- 2 Install takes about 5 min. Click Restart Now when done



- 3 When it stops on this screen, pull the USB drive, hit Enter



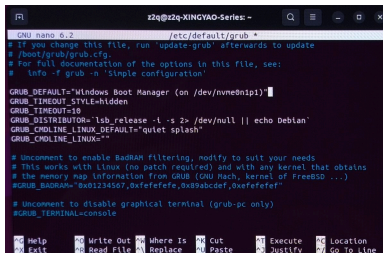
(Source: ubuntu.com)

You're in the Ubuntu desktop. See you at Week 2.

(Optional) Want Windows as the default at boot?

Dual boot defaults to Ubuntu. To switch:

- 1 Open a terminal, run `sudo nano /etc/default/grub`
- 2 Change `GRUB_DEFAULT` to `"Windows Boot Manager (on /dev/nvme0n1p1)"` (device name depends on your machine)
- 3 `Ctrl+O` to save (not `Ctrl+S`), `Ctrl+X` to exit



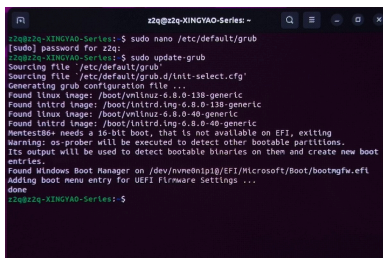
```
zq@zq-XINGYAO-Series:~$ sudo nano /etc/default/grub
# If you change this file, run `update-grub` afterwards to update
# the grub configuration.
# For full documentation of the options in this file, see:
# info -f grub -n 'Simple configuration'

GRUB_DEFAULT="Windows Boot Manager (on /dev/nvme0n1p1)"
GRUB_TIMEOUT_STYLE=hidden
GRUB_TIMEOUT=10
GRUB_DISTRIBUTOR=`lsb_release -i -s 2> /dev/null || echo Debian`
GRUB_CMDLINE_LINUX_DEFAULT="quiet splash"
GRUB_CMDLINE_LINUX=""

# Uncomment to enable BadRAM filtering, modify to suit your needs
# This works with Linux (no patch required) and with any kernel that obtains
# the memory map information from GRUB (GNU Mach, kernel of FreeBSD ...)
#GRUB_BADRAM="0x01234567,0xfefefefe,0x89abcdef,0xa4fefe"

# Uncomment to disable graphical terminal (grub-pc only)
#GRUB_TERMINAL=console
```

- 4 Run `sudo update-grub`, check the output for `Found Windows Boot Manager`



```
zq@zq-XINGYAO-Series:~$ sudo nano /etc/default/grub
zq@zq-XINGYAO-Series:~$ sudo update-grub
[sudo] password for zq:
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-6.8.0-138-generic
Found linux image: /boot/vmlinuz-6.8.0-138-generic
Found linux image: /boot/vmlinuz-6.8.0-40-generic
Found initrd image: /boot/initrd.img-6.8.0-138-generic
Found initrd image: /boot/initrd.img-6.8.0-40-generic
Warning: os-prober will be executed to detect other bootable partitions.
Its output will be used to detect bootable binaries on them and create new boot
entries.
Found Windows Boot Manager on /dev/nvme0n1p1/EFI/Microsoft/Boot/bootmgfw.efi
Adding boot menu entry for UEFI Firmware Settings ...
done
zq@zq-XINGYAO-Series:~$
```

Reboot — the GRUB menu will default to Windows Boot Manager.

References

- Stuck? Video: [8-min real-machine dual boot install \(Bilibili\)](#)
- Want to remove the dual boot later: [Uninstalling Windows10/Ubuntu dual boot \(Bilibili\)](#)
- Text version of Steps 3-4: [Installing Ubuntu dual boot without a USB drive](#)